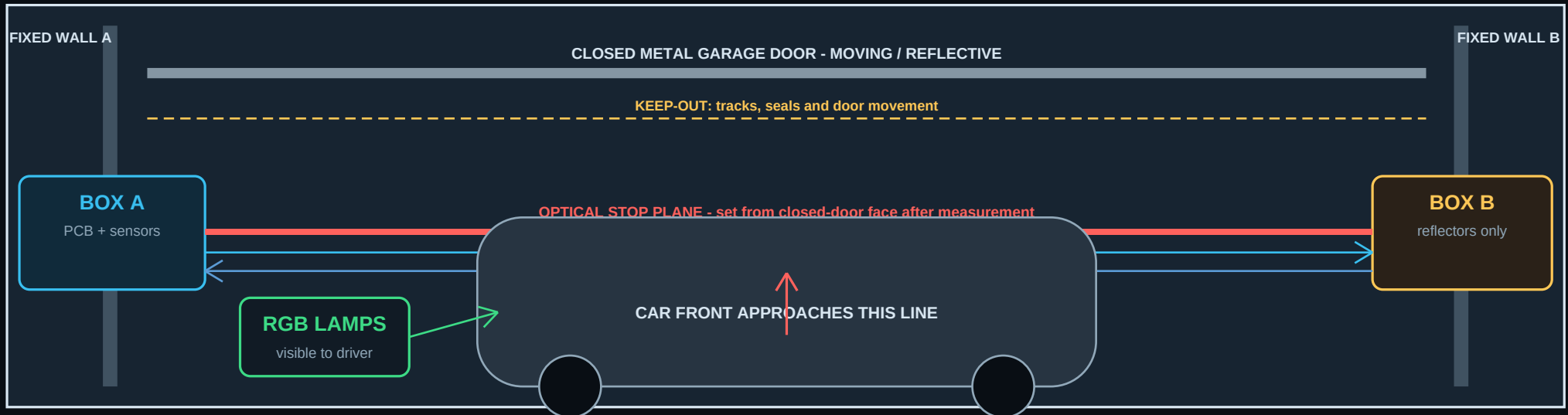
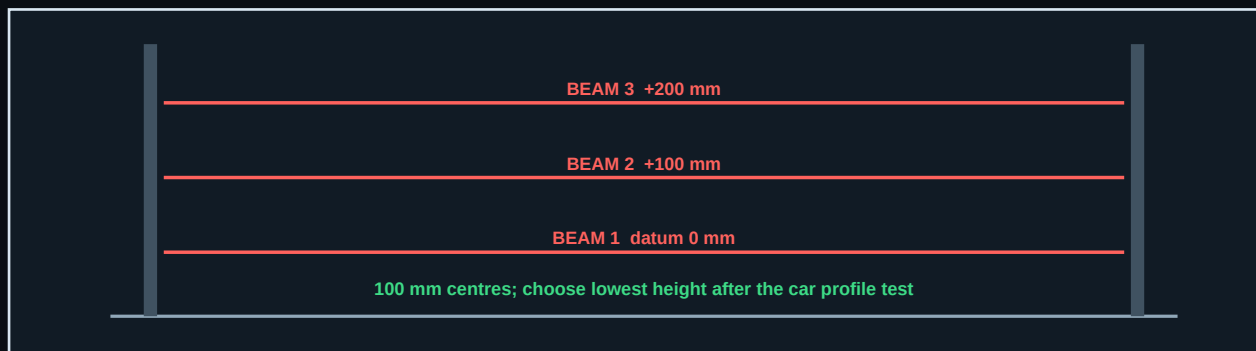


Mounting overview - fixed hardware, optical plane and driver indication

TOP VIEW - NOT TO SCALE



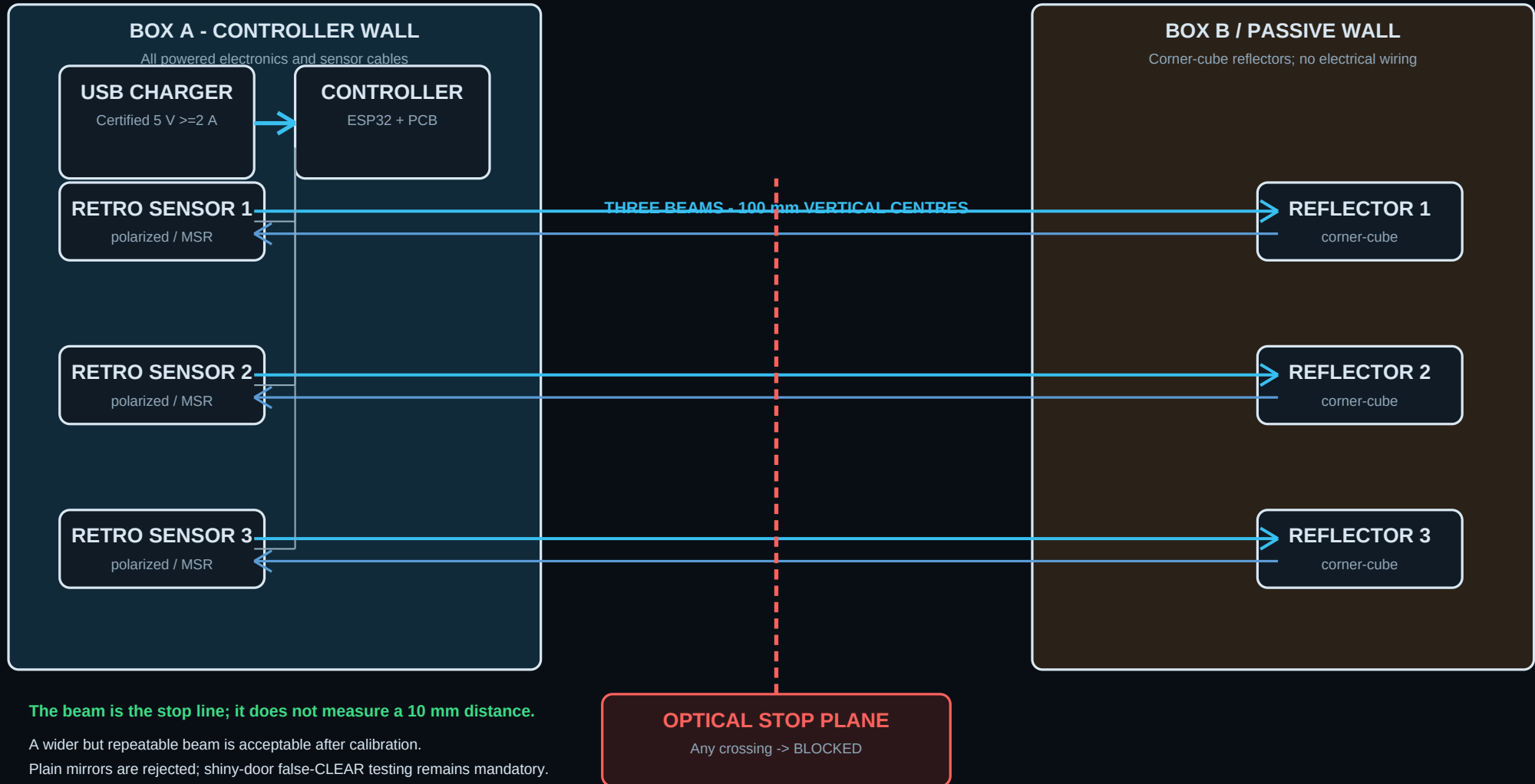
FRONT ELEVATION - EXPERIMENTAL BEAM HEIGHTS



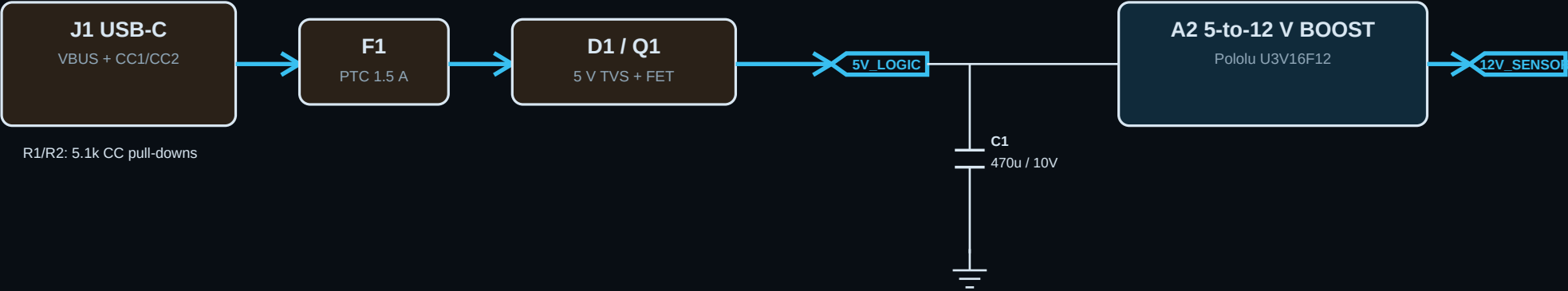
INSTALLATION RULES

1. Mount only to fixed masonry/reveal - never the door.
2. Keep optics clear of tracks, seals and moving hardware.
3. Use pan/tilt fine adjustment, then lock both axes.
4. Route the 5 V USB cable safely; add strain relief.
5. Use corner-cube reflector, not a plastic plain mirror.
6. Test shiny-door false CLEAR before trusting GREEN.
7. Measure door-to-car geometry before drilling holes.

System architecture - fixed polarized retroreflective trip lines



USB-C input, protected 5 V rail and 12 V sensor boost



POWER DESIGN NOTES

- External certified USB charger: regulated 5 V, at least 2 A.
- Plain USB-C 5 V sink; USB Power Delivery is not required.
- ESP32 uses protected 5V_LOGIC directly at VIN/5V.
- Boost converter supplies 12 V only to the three sensors.
- Never connect an industrial sensor directly to USB VBUS.
- Test points: VBUS, 5V, 12V, 3V3 and GND.



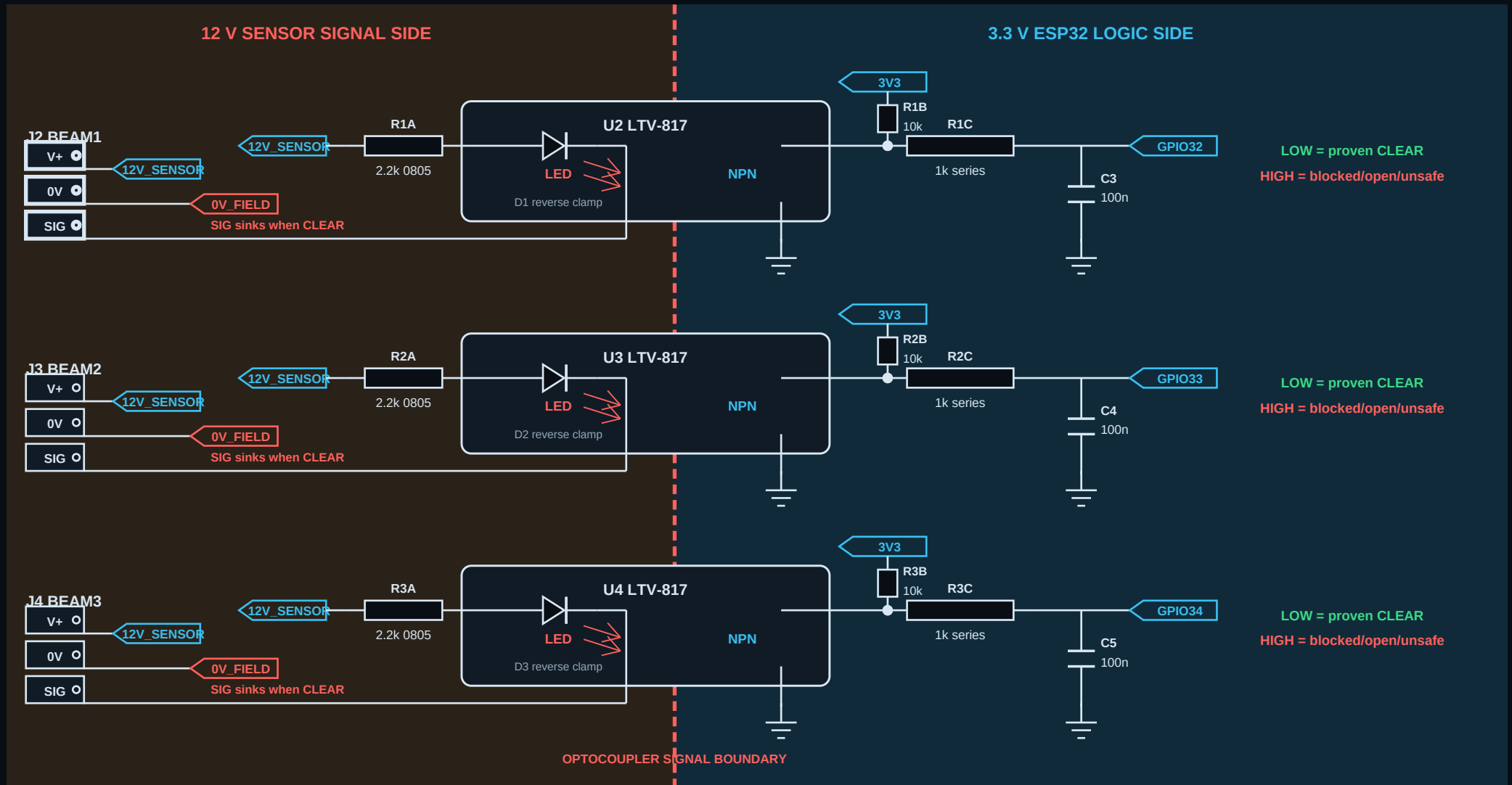
RAIL SUMMARY

5V_LOGIC: USB/ESP32 rail 12V_SENSOR: boosted field rail 3V3: logic pull-ups Parts: 0805/SOD-123/SOT-23

GARAGE BEAM SAFETY

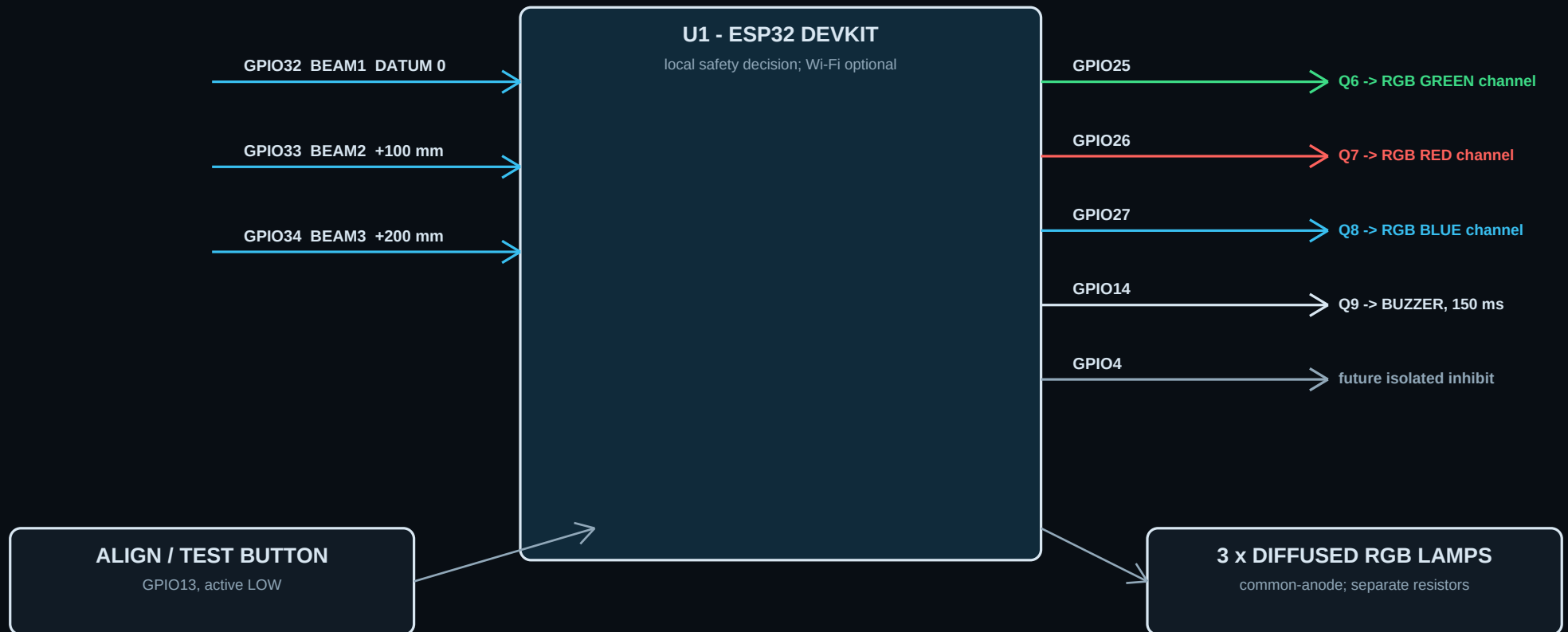
D3 THREE-BEAM CONCEPT - SHEET 4/7

Three isolated fail-safe sensor inputs at 100 mm centres



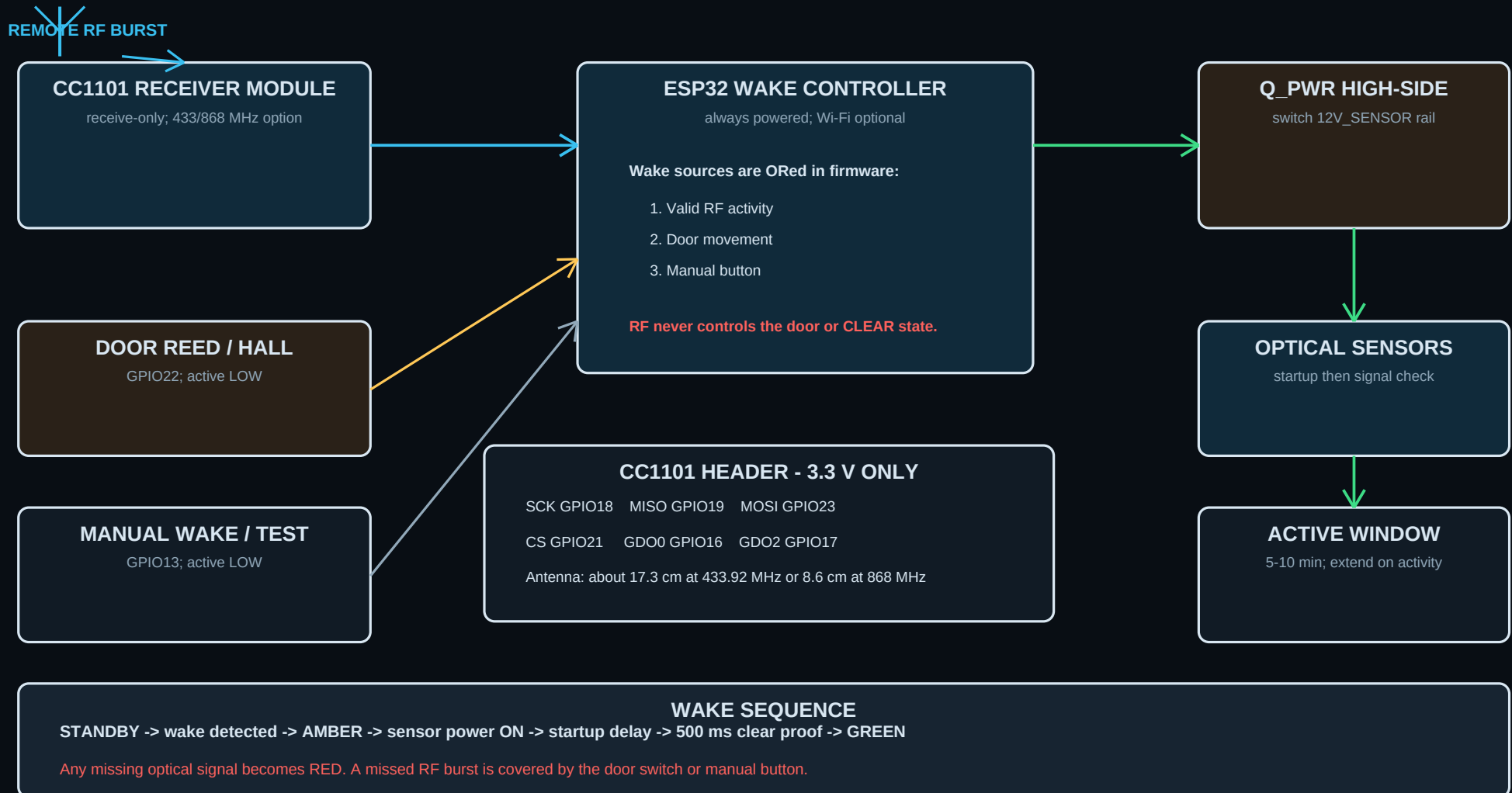
D3: all three LOW inputs are required. Blocked/open/dead sensor remains RED / NOT CLEAR.

ESP32 connections, status indication and safety behavior



GREEN=clear. RED=blocked/no signal/dead sensor. AMBER=boot/unknown (red+green channels). Dark means no power.

Optional RF, door-motion and manual wake system



D3 purchasing summary - three beams, USB-C and hand-solderable SMD

Clickable BUY/SEARCH buttons open AliExpress searches. Prices are rough delivered-EU ranges and can change.
Critical rule: the cheap optical sensor is a test article. It earns GREEN only after the repeatability and shiny-door tests.

QTY	ITEM	MINIMUM DESCRIPTION	BUDGET	LINK
3	Laser retroreflective NPN sensor	Buy one first; validate before other two	EUR 45-105	ALIEXPRESS
3	Corner-cube reflector	Matched to sensor; 100 mm centres	EUR 9-30	ALIEXPRESS
1	ESP32 DevKit V1	ESP32-WROOM, 30-pin preferred	owned / 4-9	ALIEXPRESS
1	USB-C + protected input	5.1k CC; PTC; 5V TVS; P-FET	EUR 3-8	ALIEXPRESS
1	5-to-12 V boost	Pololu U3V16F12 preferred	EUR 7-12	ALIEXPRESS
3	LTV-817S optocoupler	SMD-4; three channels	EUR 1-3	ALIEXPRESS
1	0805 resistor assortment	2.2k, 10k, 5.1k, 1k, 330R	EUR 3-8	ALIEXPRESS
1	SOD-123 diode assortment	1N4148W plus protection spares	EUR 1-4	ALIEXPRESS
1	0805 capacitor assortment	100 nF and 10 uF plus bulk caps	EUR 4-10	ALIEXPRESS

QTY	ITEM	MINIMUM DESCRIPTION	BUDGET	LINK
1	Screw terminal kit	5.08 mm, 2-way and 3-way	EUR 4-10	ALIEXPRESS
3	Diffused common-anode RGB LED	5 mm plus 3 NPN driver transistors	EUR 2-6	ALIEXPRESS
1	Active buzzer + push button	3-5 V buzzer; momentary test switch	EUR 2-5	ALIEXPRESS
1	CC1101 receiver (optional)	Match remote: 433.92 or 868 MHz	EUR 3-9	ALIEXPRESS
1	Door reed switch (recommended)	Normally closed, wired backup wake	EUR 2-6	ALIEXPRESS
1	Sensor/controller carrier	PCB 50 x 240 mm; three datums	TBD	ALIEXPRESS
1	Passive reflector carrier	PCB 30 x 240 mm; no copper	TBD	ALIEXPRESS
1	Cable glands + low-voltage cable	M12/M16 glands; 3-core sensor cable	EUR 8-20	ALIEXPRESS
2	Adjustable camera-style brackets	Metal mini pan/tilt, lockable	EUR 6-18	ALIEXPRESS

BUY LOCALLY: certified regulated USB 5 V charger rated at least 2 A. No mains voltage enters the PCB.
Use 0805, not 0402. Buy only ONE sensor/reflector pair until the optical boundary and shiny-surface tests pass.